

Computer Science 2210

Quiz 2

October 2011

Grade: 72

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1. Fill in the blanks with the **letter** corresponding to the best answer from the following list of possible answers with regard to the number of operations required for doing the indicated tasks.

A. $O(N)$ B. $O(N^2)$ C. $O(N^3)$ D. $O(\log_2 N)$ E. $O(N \log_2 N)$ F. $O(2^N)$
G. *Essentially constant time* H. *None of these*

- a) D Use a **binary search** to find a particular **Book** in a **sorted array** of **N Books**.
b) B Perform the **Sinking Sort (Bubble Sort)** to alphabetize an **array** of **N names (strings)**.
c) G Locate the alphabetically last **String** in an array of **N sorted Strings**.
d) A Locate a particular **integer** in a **List** of **N unsorted integers** if it is present.
e) A Find the largest **double** in an **List** of **N unsorted doubles**.
f) D Find a particular **String** in an array of **N sorted Strings** if it is present.
g) G Evaluate $123 - 24N + 18N^3$.
h) B Perform a **Sinking Sort** on an unsorted array of 1000 **integers**.
i) B Determine which of **N strings** in an array of **Strings** of length < 100 contains at least one space.
j) G Determine the number of items stored in a particular **List**.
k) G Compute the value of $100/N^2 + 3^N$.
l) A Find a particular **Book** object in an unsorted array of 1000 **Book** objects.
m) G Add a new **Book** object to a **List** of **N Book** objects if the **List's Capacity** exceeds its **Count**.
n) A Compute the sum of all **decimal** values in an array of **N items**.
o) A Calculate the average of **10** quiz scores for each student in a class of **N students**.
p) C Execute the following code:

```
for (int m = 1; m <= N; m++)  
    for (int n = 1; n <= 24; n++)  
        for (int k = 1; k <= N; k++)  
            Console.WriteLine ( m * n / k + N);
```

2. Consider these integer values to be sorted into **ascending** order: **53 72 16 95 39 63 24 88 44 71 57**

- a) After the first pass of a **Sinking** sort, the values are: 53 72 16 39 63 24 88 44 71 57 95
b) In the first pass, there were 11 comparisons made and there were 1 exchanges made.

3. In a sequential search of 512 items in an array, the fewest comparisons with items in the array that could be required is 1, while the maximum number is 512. If the array has been sorted into ascending order, the fewest comparisons using a binary search with items in the array that could be needed is 1, and the maximum number needed is 9.

4. The big O notation is used to describe which of these characteristics of the runtime performance of an algorithm. Circle the best answer.

- a) lower bound c) average e) least upper bound
b) upper bound d) greatest lower bound f) any of these

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5. Answer **T** or **F** indicating whether each of the following statements is **True** or **False**.

- a) T A code profiler is software
- b) F A code profiler can help identify syntax errors
- c) F A code profiler can help repair run-time program crashes
- d) T A code profiler can help identify code execution bottlenecks
- e) T A code profiler may be used to compare the run-time performance of two algorithms
- f) F Big-O notation used to describe the performance of an algorithm is determined by calculating the result of the algorithm
- g) F Big-O notation is an indication of the amount of work the algorithm must do, usually based on the number of operations performed
- h) T Big-O notation may be applied to algorithms processing small amounts of data as well as algorithms processing large amounts of data
- i) F If two algorithms are compared with **algorithm a** having **exponential** performance and **algorithm b** having **logarithmic** performance, **algorithm a** will have the best performance when the amount of data is large
- j) T An algorithm with performance $O(n \log n)$ generally performs better than an algorithm with performance $O(n^2)$ for large values of n